# Statistical Learning LAB 2

M2 EI2D & MATD – Le 13 octobre 2025

Instructions : Préparez un rapport incluant le code source et vos résultats, et déposez-le sur Teams. Pas plus de 2 personnes par groupe. N'oubliez pas de mettre les 2 noms sur le rapport, ou de faire 2 rendus. La date limite du rendu TP est le 27 octobre 2025

**Exercice 1.** *In this exercise, we study a classification problem in which the target variable $Y$ can take three different values in $\mathcal{Y} = \{A, B, C\}$. From a learning set $\mathcal{D}$, two models have been constructed $g_1$ and $g_2$. Their predictions on a new set $\mathcal{D}'$ are summarized by the following confusion matrices (we use the convention that the predicted values are in rows while the true values are in columns):*

| $g_1$ | $A$ | $B$ | $C$ |
|---|---|---|---|
| $A$ | 44 | 0 | 0 |
| $B$ | 5 | 62 | 1 |
| $C$ | 1 | 8 | 54 |

| $g_2$ | $A$ | $B$ | $C$ |
|---|---|---|---|
| $A$ | 44 | 4 | 5 |
| $B$ | 2 | 64 | 3 |
| $C$ | 4 | 2 | 47 |

*Questions :*

1. *Using the confusion matrices, compute an estimation of the distribution of $Y$, i.e. of the probabilities $P(Y = y)$ for $y \in \mathcal{Y}$.*
2. *What minimal consistency checks between $\mathcal{D}$ and $\mathcal{D}'$ should be done?*
3. *Compute the accuracy of each model on $\mathcal{D}'$ (the accuracy is the percentage of correct classification).*
4. *Determine the best model between $g_1$ and $g_2$ according to the loss function $L_b(p, t) = \mathbf{1}_{p \neq t}$. using the empirical risk on $\mathcal{D}'$.*
5. *Is the selected model the best one according to the risk associated to $L_b$ ?*
6. *We define a new loss function $\ell_2$ as follows:*

| $\ell_2(p, t)$ | $A$ | $B$ | $C$ |
|---|---|---|---|
| $A$ | 0 | 2 | 1 |
| $B$ | 1 | 0 | 1 |
| $C$ | 2 | 1 | 0 |

   *We use the convention that $p$ is the predicted value and $t$ the true value. Compute the empirical risk of each model according to this loss function on $\mathcal{D}'$.*

**Exercice 2.** *Write the Python code which allows to determine the best model between two different models $g_1$ and $g_2$ according to the loss function $L_b(p, t) = \mathbf{1}_{p \neq t}$. The two models are defined by their confusion matrices of finite dimension $m$.*